

S. aureus Eap is a polyvalent inhibitor of neutrophil serine proteases

Received for publication, June 19, 2024, and in revised form, July 23, 2024. Published, Papers in Press, August 3, 2024.
<https://doi.org/10.1016/j.jbc.2024.107627>

Nitin Mishra¹, Carson D. Gido^{1,‡}, Timothy J. Herdendorf^{1,‡}, Michal Hammel², Gregory L. Hura², Zheng-Qing Fu^{3,4}, and Brian V. Geisbrecht^{1,*}

From the ¹Department of Biochemistry & Molecular Biophysics, Kansas State University, Manhattan, Kansas, USA; ²Molecular Biophysics and Integrated Bioimaging, Lawrence Berkeley National Laboratory, Berkeley, California, USA; ³SER-CAT, Advanced Photon Source, Argonne National Laboratory, Argonne, Illinois, USA; ⁴Department of Biochemistry & Molecular Biology, University of Georgia, Athens, Georgia, USA

Reviewed by members of the JBC Editorial Board. Edited by Wolfgang Peti

Staphylococcus aureus expresses three high-affinity neutrophil serine protease (NSP) inhibitors known as the extracellular adherence protein domain (EAPs) proteins. Whereas EapH1 and EapH2 are comprised of a single EAP domain, the modular extracellular adherence protein (Eap) from *S. aureus* strain Mu50 consists of four EAP domains. We recently reported that EapH2 can simultaneously bind and inhibit cathepsin-G (CG) and neutrophil elastase (NE), which are the two most abundant NSPs. This unusual property of EapH2 arises from independent CG and NE-binding sites that lie on opposing faces of its EAP domain. Here we used X-ray crystallography and enzyme assays to show that all four individual domains of Eap (*i.e.* Eap1, Eap2, Eap3, and Eap4) exhibit an EapH2-like ability to form ternary complexes with CG and NE that inhibit both enzymes simultaneously. We found that Eap1, Eap2, and Eap3 have similar functional profiles insofar as NSP inhibition is concerned but that Eap4 displays an unexpected ability to inhibit two NE enzymes simultaneously. Using X-ray crystallography, we determined that this second NE-binding site in Eap4 arises through the same region of its EAP domain that also comprises its CG-binding site. Interestingly, small angle X-ray scattering data showed that stable tail-to-tail dimers of the NE/Eap4/NE ternary complex exist in solution. This arrangement is compatible with NSP-binding at all available sites in a two-domain fragment of Eap. Together, our work implies that Eap is a polyvalent inhibitor of NSPs. It also raises the possibility that higher-order structures of NSP-bound Eap may have unique functional properties.

The complement/neutrophil axis serves an early and critical role in the innate immune response to bacterial pathogens. Human clinical evidence shows that persons with deficiencies in either complement components (1) or circulating neutrophils (2) experience an increased incidence of serious bacterial infections when compared to healthy individuals. The widespread evolution of bacterial evasion strategies targeting the complement/neutrophil axis can likewise be considered

evidence of the importance of these systems to anti-bacterial immunity (3, 4). The Gram-positive bacterium *Staphylococcus aureus* is particularly active in its evasion of the complement/neutrophil axis, as it expresses over two dozen unique proteins that target components of complement and neutrophils (5, 6). Collectively, these immune evasion proteins disrupt opsonization by complement fragments, neutrophil chemotaxis, phagocytosis, and subsequent killing with the neutrophil phagosome (5, 6). Study of these proteins at the molecular level has identified several new modes of complement inhibition (7) and an inhibitor of the HOCl-generating myeloperoxidase within neutrophils (8). It has also identified a new family of highly selective inhibitors of the neutrophil serine proteases (NSP), neutrophil elastase (NE), cathepsin-G (CG), and proteinase-3 (PR3).

The EAP family is comprised of three unique secreted proteins produced by *S. aureus* (9, 10; Fig. S1). Collectively, they are defined by an archetypal ~100-residue β -grasp domain (*i.e.* EAP domain) consisting of a large α -helix lying diagonally across a mixed β -sheet with five long β -strands (9). This domain occurs in either single or repeating units among the family members and ranges from 26% to 83% amino acid identity in its various instances within *S. aureus* strain Mu50 (9). EapH1 (NCBI reference sequence: WP_000769705) and EapH2 (NCBI reference sequence: WP_000769694) each contain a single EAP domain, while the larger extracellular adherence protein (Eap; NCBI reference sequence: WP_001557458) contains four such domains in the variant produced by *S. aureus* strain Mu50 (9). Other *S. aureus* isolates express forms of Eap that consist of five or six EAP domains (10), although the functional significance of such differences has not been established. At a structural level, EAP domains display a significant level of homology to the well-studied superantigen-class toxins produced by Gram-positive cocci (9). While this unexpected finding led to the hypothesis that EAP proteins might likewise serve a role in immune evasion (9), it took nearly a decade longer until Stapels *et al.* showed that EAP proteins act as selective inhibitors of NSPs (11). These authors found that all three EAP proteins exhibit dose-dependent inhibition of NE, CG, and PR3 and that they seemed to have at least partly overlapping activities (11).

[‡] These authors contributed equally to this work.

* For correspondence: Brian V. Geisbrecht, geisbrechtb@ksu.edu.

Stapels *et al.* subsequently found that an *S. aureus* strain deleted for the gene encoding Eap was statistically indistinguishable from an isogenic control in a mouse liver abscess model of infection, but that a corresponding strain deleted for all three EAP encoding genes was $\sim 10^2$ -fold less infectious (11). Although these observations were broadly consistent with the concept of functional redundancy, they offered little insight as to why *S. aureus* produces three different NSP inhibitors. They also did not address whether fundamental mechanistic differences exist between these EAP proteins insofar as NSP inhibition is concerned.

EAP family members do not share obvious sequence or structural properties with other known protease inhibitors (11). Therefore, we have been systematically studying their structure/function relationships to better understand this staphylococcal immune evasion protein family. Our initial crystallographic studies on CG (12) and NE (11) inhibited by EapH1 revealed that this EAP protein uses a globally similar binding mode to block the activity of both proteases. This binding mode involves the loop connecting $\beta 1$ and $\beta 2$ of EapH1, as well as the C-terminal region of the prominent α -helix that lies across its mixed β -sheet (9, 11, 12). EapH1-R89 lies at the C-terminal end of this α -helix, and site-directed mutagenesis studies showed that its contributions are paramount to the inhibition of both CG (12) and NE (13) by EapH1. Because EapH2 is 43% identical to EapH1 at the amino acid level, we predicted that these two proteins would follow similar structure/function relationships (14). Our initial studies based on molecular modeling did not support this hypothesis (15), however, and further site-directed mutagenesis analysis of EapH2-R90 (which is equivalent to EapH1-R89) showed that this residue is not essential to NE inhibition by EapH2 (16). Paradoxically, we found that EapH2-R90 contributes significantly to CG inhibition by EapH2 (17).

Since we were unable to crystallize either CG or NE bound to EapH2, we used NMR chemical shift perturbation studies to explore the NSP-binding sites of EapH2 and included EapH1 as a control (17). The EapH1 binding sites for both CG and NE identified through NMR were consistent with previous cocrystal structures. Still, the results for EapH2 revealed something unexpected: whereas the CG-binding site of EapH2 involved a globally similar binding mode to EapH1, the NE-binding site of EapH2 involved an entirely different region of the inhibitor spanning most of the residues between $\beta 5$ and $\beta 8$ (17). Because this region lies on the opposite side of the EAP domain as the conventional NSP-binding site, these observations suggested that EapH2 might have the unusual ability to bind and inhibit two different NSPs simultaneously. Thereafter, we determined several crystal structures of EapH2 bound in ternary complex with both CG and NE, and we showed using enzyme assays that both enzymes were indeed inhibited (17). Together, these studies defined a new mode of simultaneous inhibition of two different serine proteases by this EAP family member.

The observations described above indicate a surprising level of structure/function diversity among EAP family members. Considering that, the multidomain nature of the Eap fusion

protein has raised additional questions. Do the individual domains within Eap utilize a single, overlapping site to inhibit NSPs like EapH1? Or are they instead capable of binding two NSPs at different sites to affect simultaneous inhibition like EapH2? If the latter is true, is their ability to bind two NSPs simultaneously influenced by linking the domains into a fusion protein? Finally, since both EapH1 and EapH2 contain only a single EAP domain yet display potent NSP inhibitory properties (11–13, 16, 17), what purpose might be served by linking individual EAP domains together into a larger fusion protein? To address these questions, we recently carried out a structure/function study of the first (*i.e.* Eap1) and second (*i.e.* Eap2) domains of Eap from *S. aureus* strain Mu50 (18). This work showed that both Eap1 and Eap2 inhibit CG using an overall binding mode like that used by both EapH1 and EapH2 (12, 17) but that they inhibit NE using an alternative binding mode comparable to that used by EapH2 (17). An important implication of this work is that Eap1 and Eap2 should be capable of binding and inhibiting CG and NE simultaneously, much like EapH2 (17). However, this has not yet been shown experimentally. Furthermore, the structure/function properties of Eap3 and Eap4 as NSP inhibitors remain unknown. These knowledge gaps have, therefore, been a significant impediment to answering the questions posed above.

In this report, we used X-ray crystallography and functional assays to determine that all four domains of Eap can simultaneously inhibit CG and NE. Through detailed enzyme kinetic investigations and supporting X-ray crystallography, we further found that Eap4 is uniquely able to bind and inhibit two copies of NE simultaneously. Thereafter, we used small angle X-ray scattering analyses (SAXS) to show that ternary complexes of Eap4 with NE and/or CG form stable dimers in solution that result from NSP binding. Finally, we used SAXS to show that the NSP-induced dimerization of Eap4 is maintained in a two-domain fragment of Eap and in a manner that supports the binding of NSPs at all available sites. Our work supports a model wherein the full-length Eap protein is a polyvalent NSP inhibitor that can assemble into higher-order structures. Some potential biological ramifications of these observations are discussed.

Results

Individual subdomains of Eap can simultaneously bind CG and NE

EapH2 contains two independent NSP-binding modes and is capable of simultaneously inhibiting both CG and NE (17). Since Eap1 and Eap2 each contain independent CG- and NE-binding sites similar to those in EapH2 (18), we predicted that these EAP domains might also be capable of binding and inhibiting CG and NE simultaneously. We solved and refined crystal structures of ternary CG- and NE-bound complexes for Eap1 (*i.e.* CG/Eap1/NE) and Eap2 (*i.e.* CG/Eap2/NE) at 3.63 Å and 3.10 Å limiting resolution, respectively (Figs. 1, A and B, S2, A–F, and Table 1). Thereafter, we solved and refined crystal structures of ternary CG- and NE-bound complexes for Eap3 (*i.e.* CG/Eap3/NE) and Eap4 (*i.e.* CG/Eap4/NE) at 1.95 Å

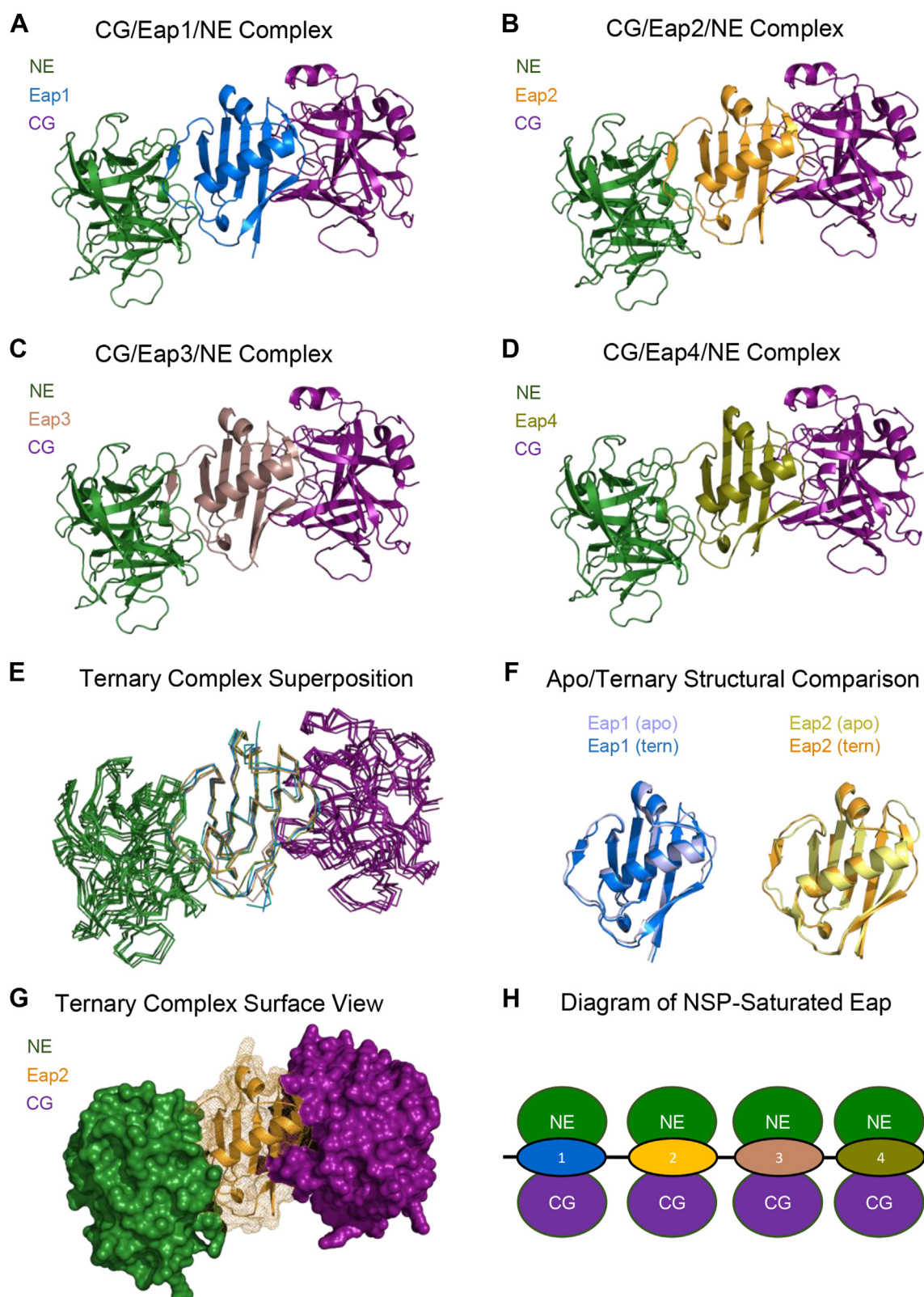


Figure 1. Ternary complexes of EAP domains bound to CG and NE define a structural basis for bifunctional inhibition of neutrophil serine proteases. *A*, refined model for an individual ternary complex comprised of CG (purple), Eap1 (light blue), and NE (dark green) drawn as a ribbon diagram. *B*, refined model for an individual ternary complex comprised of CG (purple), Eap2 (light orange), and NE (dark green) drawn as a ribbon diagram. *C*, refined model for the ternary complex comprised of CG (purple), Eap3 (bronze), and NE (dark green). *D*, refined model for an individual ternary complex comprised of CG (purple), Eap4 (gold), and NE (dark green). *E*, superposition of representative ternary complexes for various EAP domains with CG (purple) and NE (dark green). Individual EAP domains are colored as in the panels above, while EapH2 is colored teal. Structures are presented as α wire diagrams for clarity. *F*, comparison of the structures of Eap1 and Eap2 in their free and CG/NE-bound forms. Superpositions of ribbon diagrams for free Eap1 (lavender) and bound Eap1 (light blue) are shown alongside those of free Eap2 (light yellow) and bound Eap2 (light orange). Note there is little rearrangement within either

Table 1
X-ray diffraction data collection, structure solution, and refinement statistics^a

Structure	CG/Eap1/NE 8D7I	CG/Eap2/NE 8D7K	CG/Eap3/NE 9ASX	NE/Eap4 9ASS	CG/Eap4/NE 9ATK	NE/Eap4/NE 9ATU
Data Collection						
Beamline	APS 22-ID	APS 22-ID	APS 22-ID	APS 22-ID	APS 22-ID	APS 22-ID
Space Group	$P2_12_12_1$	$P2_1$	$P2_12_12_1$	$F222$	$P1$	$P2_12_12_1$
Wavelength (Å)	1.000	1.000	1.000	1.000	1.000	1.000
Cell Dimensions						
a, b, c (Å)	102.23	97.59	43.97	94.62	82.21	75.82
	152.27	107.44	110.39	127.38	85.53	121.06
	296.38	105.42	116.94	148.68	89.54	147.63
α, β, γ (°)	90.00	90.00	90.00	90.00	82.77	90.00
	90.00	91.52	90.00	90.00	85.23	90.00
	90.00	90.00	90.00	90.00	83.60	90.00
Resolution (Å)	100.00–3.63	50.00–3.10	50.00–1.95	50.00–1.75	50.00–2.10	50.00–2.05
Wilson B-Factor (Å ²)	84.5	51.0	24.3	23.4	30.8	43.0
Completeness (%)	99.9 (100.0)	99.1 (99.8)	99.5 (98.9)	99.4 (97.8)	90.4 (87.0)	99.7 (99.3)
I/ σ I	14.3 (4.7)	10.2 (2.8)	10.9 (3.6)	11.5 (1.8)	9.4 (1.7)	20.3 (1.5)
R _{pim}	0.168 (0.768) ^b	0.071 (0.343)	0.111 (0.739)	0.064 (0.641)	0.111 (0.432)	0.029 (0.442)
CC _{1/2}	0.998 (0.938)	0.983 (0.762)	0.965 (0.442)	0.809 (0.580)	0.966 (0.386)	0.990 (0.600)
Redundancy	9.8 (10.18)	4.8 (5.0)	10.4 (6.5)	10.0 (5.5)	2.8 (2.1)	12.8 (9.8)
Refinement						
Resolution (Å)	96.64–3.63	43.83–3.10	40.14–1.95	41.51–1.75	47.89–2.10	41.28–2.05
Number of reflections	51,218	38,170	41,557	44,362	117,167	79,893
R _{work} /R _{free} (%)	18.6/21.7	18.4/25.8	19.3/22.8	18.9/21.2	17.9/23.3	18.5/23.7
Atoms modeled						
Total	25,188	16,684	4519	2751	17,992	8816
Water	0	0	205	246	770	338
Ion	0	0	0	1	0	0
Ramachandran plot						
Favored/allowed (%)	93.70/5.58	94.27/5.31	96.44/3.56	98.08/1.92	97.23/2.72	96.69/3.31
B-Factors (Å ²)	95.0	46.0	29.0	32.0	41.0	55.0
R.M.S. deviations						
Bond lengths (Å)	0.003	0.009	0.007	0.007	0.008	0.008
Bond angles (°)	0.69	1.10	0.86	0.80	0.91	0.89

^a Values in parentheses are for the highest-resolution shell.^b Values represent R_{merge} as calculated by xScale.

and 2.10 Å limiting resolution, respectively (Figs. 1, C and D, S2, G–L, and Table 1). Aside from the Eap3 ternary complex, the crystallographic asymmetric units for the Eap1, Eap2, and Eap4 ternary complexes each contained multiple copies of their respective biological units (Fig. S2, A, D, G, and J); however, the models for these individual biological units from the same crystal superimposed well upon each other (*Data Not Shown*), and representative models of each EAP domain ternary complex shared clear positional similarities with one another and with CG/EapH2/NE (Fig. 1E). There was also well-defined electron density encompassing the entire ternary complex in all structures (Fig. S2, B, E, H, and K). Finally, there was a relatively uniform distribution of B-factors within the EAP domains of representative ternary complex models (Fig. S2, C, F, I, and L). This most likely resulted from protein–protein interactions that limit molecular motions in regions that are otherwise dynamic without a binding partner, as was seen in the structures of Eap1 and Eap2 bound individually to CG and NE (18). Although the lack of a complete portfolio of apo or binary NSP-bound structures for Eap3 and Eap4 precluded making a conclusive statement for these EAP domains, the similarities in B-factor distribution between the Eap1 and Eap2 binary complexes and that of NE/Eap4 suggested that

this was a general feature of these structures (Fig. S3 and Table 1). Therefore, we conclude that all EAP domains are likely to exhibit conformational flexibility at their respective NSP-binding sites in the absence of ligand.

Because we previously reported crystal structures of Eap1 and Eap2 in their unbound forms (9, 18) and bound in binary complexes to CG and NE (18), we compared the structural features of these two EAP domains in their free, binary, and ternary complex states. We found the structures of the Eap1 and Eap2 ternary complexes were an amalgamation of their respective binary complexes with NSPs. In each case, CG occupied the EapH1-like binding site seen in the CG/Eap1 and CG/Eap2 structures (18), while NE occupied the EapH2-like binding site seen in the NE/Eap1 and NE/Eap2 structures (18). Superposition of the EAP domain structures from the ternary complexes with their unbound forms revealed significant positional rearrangement within both the CG- and NE-binding sites (Fig. 1F). This resulted in what appeared to be new stretches of β -strand in both NSP-binding sites, which arose *via* the formation of multiple hydrogen bonds between mainchain-derived groups from the inhibitor and each protease. While these hydrogen bonding networks were extensive and included interactions between the EAP domains and

inhibitor, aside from the loop regions that comprise most of the protease-binding sites. G, representative image of a CG/EAP/NE ternary complex drawn as a molecular surface. A refined model of the CG/Eap2/NE complex is shown with NE and CG as solid surfaces, while Eap2 is presented as a semitransparent mesh. Roughly 33% (Eap1), 36% (Eap2), 34% (Eap3), and 32% (Eap4) of the ~6100 Å² total surface area available in the EAP domain is buried upon ternary complex formation. The N- and C-termini of the EAP domain are at the bottom and top of the image, respectively. H, schematic representation of full-length Eap (colored ovals) saturated with CG (purple circles) and NE (dark green circles).

groups derived from the catalytic serine of both proteases, comparable interactions were also seen in the binary NSP-bound complexes of Eap1 and Eap2 (18). Likewise, the shape complementarity (s.c.) coefficients and total buried surface areas of the individual interfaces within the ternary complexes compared favorably to the binary complexes of these EAP domains (Table S1): CG/Eap1/NE buried a total of 2028 Å², which was ~5% greater than the sum of CG/Eap1 and NE/Eap1 (*i.e.* 1937 Å²), while CG/Eap2/NE buried a total of 2128 Å², which was ~10% greater than the sum of CG/Eap2 and NE/Eap2 (*i.e.* 1931 Å²). These buried surface area values corresponded to ~33% and ~36% of the total available surface area in Eap1 and Eap2, which was most easily visualized by examining a ternary complex as a molecular surface (Fig. 1G).

We felt that the N- and C-termini lying at opposing sides of the EAP domains from each other was potentially significant. Indeed, Eap1, Eap2, Eap3, and Eap4 do not exist as free-standing proteins *per se* but as part of a multidomain fusion protein known as Eap (9, 19). In this regard, we previously reported that the four-domain Eap from *S. aureus* strain Mu50 adopted an extended structure in solution (19). This structure provided considerable space between adjacent subdomains and further suggested that Eap may form ternary complexes like those described above at NSP at all its available sites (Fig. 1H). But since the structure/function properties of Eap 3 and Eap4 as NSP inhibitors had not yet been studied, we first sought additional information in that area to ensure that any unanticipated differences would not be overlooked.

All individual domains of full-length Eap simultaneously inhibit CG and NE

The ternary complexes presented above shared significant similarities to that of CG/EapH2/NE (17). Although these structures strongly suggested that each EAP domain could simultaneously inhibit both CG and NE like EapH2 (17), we sought experimental evidence to support this assumption. To test the ability of the individual EAP domains to inhibit CG and NE simultaneously, we compared the progress curves and extent of inhibition of reactions containing both NSP enzymes, their substrates, and an EAP domain inhibitor with reactions consisting of only one NSP, both protease substrates, and the same EAP domain inhibitor (Fig. 2). To distinguish between the different NSP reactions, we employed a colorimetric substrate of CG (*i.e.* succinyl-Ala-Ala-Pro-Phe-p-nitroanalide, SA₂PF-pNA) and a fluorescent substrate of NE (*i.e.* methoxysuccinyl-Ala-Ala-Pro-Val-7-amino-4-methylcoumarin, MSA₂PV-AMC). We found that the extent of CG inhibition upon concurrent exposure of the enzyme to its substrate and the individual EAP domains was unchanged by the presence of NE (Fig. 2, A, C, E, and G). Similarly, we found that the extent of NE inhibition upon concurrent exposure of the enzyme to its substrate and the individual EAP domains was unchanged by the presence of CG (Fig. 2, B, D, F, and H). The simplest interpretation of these data was that the presence of NE did not influence the inhibitory properties of the CG site within these EAP domains and *vice versa*. Together, these results provided evidence that Eap1,

Eap2, Eap3, and Eap4 can each simultaneously bind and inhibit both CG and NE in solution, consistent with the structural observations presented above (Fig. 1).

Characterization of NSP inhibition by Eap3 and Eap4 identifies divergent functional properties of Eap4

EapH1 exhibits reversible, time-dependent inhibition of both CG and NE through a single, high-affinity NSP-binding mode (11–13). Although EapH2 largely shares these functional properties, structural and mutagenesis studies revealed that EapH2 contains two distinct NSP-binding sites to inhibit CG and NE (16, 17). Eap1 and Eap2 likewise contain equivalent CG- and NE-binding sites to EapH2 (17, 18). Evaluation of residues found within these sites by site-directed mutagenesis revealed certain trends: whereas the CG-binding site of all three inhibitors was centered on a conserved arginine found at the end of the prominent α -helix of their EAP domain (*i.e.* EapH2-R90, Eap1-R93, and Eap2-R202), their NE-binding site was predicated upon residues (*i.e.* EapH2-D122, Eap1-T128, and Eap2-T237/N239) found on the opposite side of their EAP domains (Fig. S1). Because information on NSP binding by Eap3 and Eap4 was unavailable till now (Figs. 1, C and D and S4), we sought to evaluate our structural observations through a combination of site-directed mutagenesis and functional assays of enzyme inhibition. Since all EAP domains examined thus far exhibited time-dependent inhibition (12, 13, 16–18), we used global fitting of reaction progress curves to analyze our data. This allowed insight into whether the association or dissociation rate constants relevant to each complex were affected by the loss of key residues (20, 21).

We initially focused on inhibition of CG by Eap3 and Eap4 (Table 2). We found that WT Eap3 inhibited CG in a time-dependent manner with an apparent inhibition constant (K_i) of 44 nM. Mutation of the conserved arginine to methionine (*i.e.* Eap3-R311M) resulted in an inflation in the apparent K_i by ~6.5-fold to 284 nM; this effect was due to an increase in the off-rate of Eap3 (*i.e.* first-order rate constant k_5 ; Table 2). This observation was consistent with Eap3 inhibiting CG in an EapH2-like manner as seen in the ternary complex structure reported above (Figs. 1C and S4A). Interestingly, we found that WT Eap4 was an outlier among EAP domains in terms of its CG inhibition properties. Unlike other WT EAP domains, Eap4 did not inhibit CG in a time-dependent manner and had a much higher apparent K_i of 434 nM. Although mutation of the conserved arginine to methionine (*i.e.* Eap4-R416M) resulted in a modest ~2.4-fold inflation of the K_i to 1050 nM, this observation was still consistent with Eap4 inhibiting CG in an EapH2-like manner as seen in the ternary complex structure (Figs. 1D and S4B). Aside from this noteworthy decrease in potency, the CG inhibition properties of Eap3 and Eap4 shared many features in common with Eap1 and Eap2, as well as EapH2 (17, 18).

We then turned our attention to NE inhibition by Eap3 and Eap4 (Table 3). We found that WT Eap3 inhibited NE in a time-dependent manner with an apparent inhibition constant (K_i) of 3.5 nM. Mutation of two positions corresponding to the residues important for NE inhibition by EapH2

Polyvalent inhibition of neutrophil serine proteases by Eap

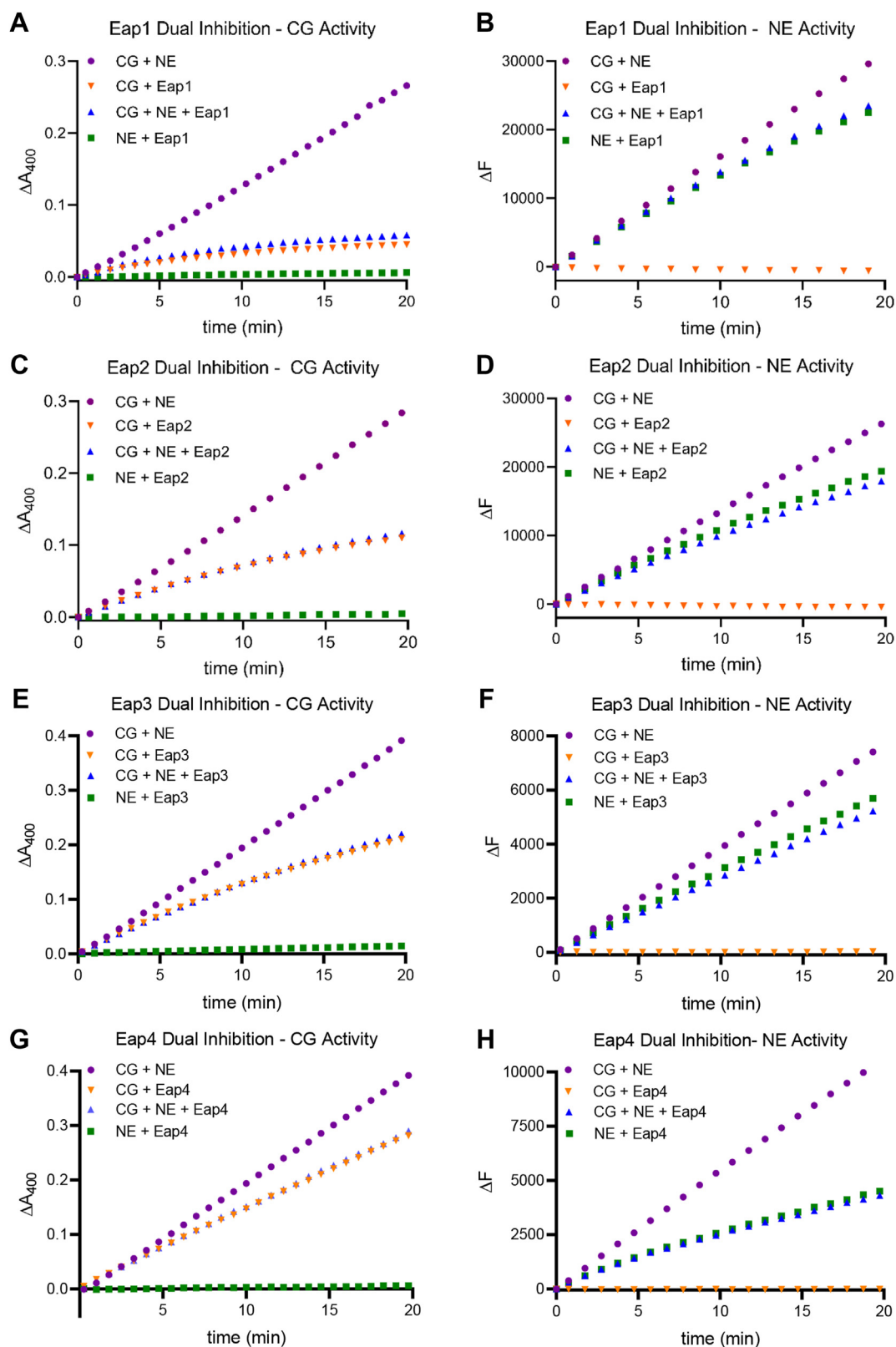


Figure 2. Simultaneous inhibition of CG and NE by individual domains of Eap. The abilities of EAP domains to inhibit CG and NE simultaneously were followed by pairwise reactions that investigated either CG (colorimetric) or NE activity (fluorometric). The substrate concentrations in all reactions were 1.175 mM SA₂PF-pNA and 170 μM MSA₂PV-AMC. A, simultaneous inhibition of CG and NE by Eap1 followed by SA₂PF-pNA turnover by CG. (●) Reaction consists of 7.5 nM CG, 7.6 nM NE, 0 nM Eap1. (▼) Reaction consists of 7.5 nM CG, 0 nM NE, 9.4 nM Eap1. (▲) Reaction consists of 7.5 nM CG, 7.6 nM NE, 9.4 nM Eap1. (■) Reaction consists of 0 nM CG, 7.6 nM NE, 9.4 nM Eap1. B, simultaneous inhibition of CG and NE by Eap1 followed by MSA₂PV-AMC turnover by NE. (●) Reaction consists of 7.5 nM CG, 7.6 nM NE, 0 nM Eap1. (▼) Reaction consists of 7.5 nM CG, 0 nM NE, 9.4 nM Eap1. (▲) Reaction consists of 7.5 nM CG, 7.6 nM NE, 9.4 nM Eap1. (■) Reaction consists of 0 nM CG, 7.6 nM NE, 9.4 nM Eap1. C, simultaneous inhibition of CG and NE by Eap2 followed by SA₂PF-pNA turnover by CG. (●) Reaction consists of 8.2 nM CG, 7.3 nM NE, 0 nM Eap2. (▼) Reaction consists of 8.2 nM CG, 0 nM NE, 10.3 nM Eap2. (▲) Reaction

Table 2Kinetic rate constant estimates of cathepsin G inhibition determined by progress curve analysis^a

Inhibitor	k_f ($M^{-1}s^{-1}$)	SD	k_3 (s^{-1})	SD	k_d ($M^{-1}s^{-1}$)	SD	k_s (s^{-1})	SD	K_i (nM)	SD
Eap3 WT	2.7×10^3	1.3×10^3	7.2	2.3	8.3×10^3	1.0×10^3	3.7×10^{-4}	1.1×10^{-4}	44	13
Eap3 R311M	3.9×10^3	1.2×10^3	9.4	1.6	1.1×10^4	1.9×10^3	3.1×10^{-3}	5.2×10^{-4}	284	3.8
Eap4 WT ^b	-	-	11.2	1.4	-	-	-	-	434	35
Eap4 R416M ^b	-	-	13.0	2.8	-	-	-	-	1050	95

^a Microscopic rate constant estimates are the average of at least three individual sets of 30 progress curves (two [CG], three [substrate] per [CG], and five [inhibitor] per [substrate]).^b Due to the loss of the time-dependent inhibition characteristic, mutants were analyzed by a classical 4 × 5 inhibition assay. K_i estimated by globally fitting three assay sets. Error is the standard error of the global fit.

(i.e. Eap3-T346A and Eap3-N348A) resulted in an ~12.8-fold and ~11.6-fold increase in the apparent K_i to 44.7 nM and 40.7 nM, respectively; this effect was predominately due to an increase of the off-rate of Eap3 (i.e. first-order rate constant k_5 ; Table 3). The double mutant (i.e. Eap3-T346A/N348A) displayed an even larger ~152-fold increase in apparent K_i to 532 nM, as well as a change from time-dependent to more classical inhibition. These observations were consistent with Eap3 inhibiting NE in an EapH2-like manner as seen in the structure above (Figs. 1C and S4C).

In contrast to its effects on CG, we found that WT Eap4 inhibited NE relatively tightly with an apparent K_i of 0.27 nM (Table 3). Eap4 displayed an unexpected ability to potently inhibit NE at substoichiometric concentrations compared to other EAP domains, which strongly suggested the existence of multiple NE-binding sites within this inhibitor (n.b.: this subject is explored in the following subsection). Similarly to Eap3, mutation of the conserved arginine to methionine (i.e. Eap4-R416M) did not appreciably change its properties compared to WT. However, mutation of the equivalent threonine and asparagine to alanine to those examined in Eap3 (i.e. Eap4-T451A and Eap4-N453A) resulted in an ~93-fold and ~47-fold increase in apparent K_i to 25.2 nM and 12.7 nM, respectively; in contrast to the observations for Eap3, the double mutant (i.e. Eap4-T451A/N453A) inhibited NE similarly to Eap4-T451A. Each of these mutations also resulted in loss of the time-dependent inhibitory characteristic. Interestingly, mutation of the proline between these two positions to alanine (i.e. Eap4-P452A), either alone or in combination with the other mutations, resulted in WT to only moderately perturbed inhibition (<15-fold). Examination of the interface between Eap4 and NE revealed that the interactions were governed by hydrogen bonds between the polypeptide backbone of the inhibitor and protease (Figs. 1D and S4D). This could partly account for the unusual effects resulting from the loss of Eap4-P452.

Eap4 forms ternary inhibitory complexes with NE

The observations above revealed that Eap4 was an outlier among this family of inhibitors for two reasons. First, it inhibited CG ~10-fold weaker than any other EAP domain investigated thus far ((12, 17, 18) and Table 2). Second, it displayed an apparent ability to inhibit NE at substoichiometric concentrations to a far greater extent than expected. We reasoned that this latter feature could be explained most easily by multiple NE-binding sites on Eap4. To gain deeper insight into this unexpected possibility, we prepared samples of Eap4 in the presence of two molar equivalents of NE, obtained single crystals, and solved and refined a structure of the NE/Eap4/NE ternary complex to 2.05 Å limiting resolution (Figs. 3, A and B, S5, A and B, and Table 1). We found that Eap4 bound one NE at the same site seen in the binary NE/Eap4 and ternary CG/EAP/NE complexes ((17); Figs. 1, S2, and S3), which we denote hereafter as the “conserved site,” while it bound to a second NE at the CG-binding site seen in the ternary CG/EAP/NE complexes ((17); Figs. 1 and S2), which we denote hereafter as the “alternative site.” Interestingly, this alternative site resembled the overall NE-binding mode of EapH1 (11). Eap4 interacted with the catalytic serine of the NE at each respective binding site (Fig. S5, C and D). While the presence of this second NE-binding site appeared to be unique among EAP domains, the nature and extent of the interactions found within both the conserved and alternative NE-binding sites (Table S1), as well as the distribution of B-factors (Fig. S5E), were comparable to those described for other EAP domains either previously or earlier in this report ((11, 17, 18); Fig. 1).

With the NE/Eap4/NE structure as a guide, we attempted to isolate the two NE-binding sites for functional analysis (Table 3). We created individual proline to cysteine mutants within either the alternative (i.e. Eap4-P386C) or conserved site (i.e. Eap4-P452C). Then, we crosslinked these individual mutant proteins using 1,11-bismaleimido-triethyleneglycol

consists of 8.2 nM CG, 7.3 nM NE, 10.3 nM Eap2. (■) Reaction consists of 0 nM CG, 7.3 nM NE, 10.3 nM Eap2. D, simultaneous inhibition of CG and NE by Eap2 followed by MSA₂PV-AMC turnover by NE. (●) Reaction consists of 8.2 nM CG, 7.3 nM NE, 0 nM Eap2. (▼) Reaction consists of 8.2 nM CG, 0 nM NE, 10.3 nM Eap2. (▲) Reaction consists of 8.2 nM CG, 7.3 nM NE, 10.3 nM Eap2. (■) Reaction consists of 0 nM CG, 7.3 nM NE, 10.3 nM Eap2. E, simultaneous inhibition of CG and NE by Eap3 followed by SA₂PF-pNA turnover by CG. (●) Reaction consists of 17.7 nM CG, 15.4 nM NE, 0 nM Eap3. (▼) Reaction consists of 17.7 nM CG, 0 nM NE, 38.6 nM Eap3. (▲) Reaction consists of 17.7 nM CG, 15.4 nM NE, 38.6 nM Eap3. (■) Reaction consists of 0 nM CG, 15.4 nM NE, 38.6 nM Eap3. F, simultaneous inhibition of CG and NE by Eap3 followed by MSA₂PV-AMC turnover by NE. (●) Reaction consists of 17.7 nM CG, 15.4 nM NE, 0 nM Eap3. (▼) Reaction consists of 17.7 nM CG, 0 nM NE, 17.7 nM Eap4. (▲) Reaction consists of 17.7 nM CG, 15.4 nM NE, 17.7 nM Eap4. (■) Reaction consists of 0 nM CG, 15.4 nM NE, 17.7 nM Eap4. H, simultaneous inhibition of CG and NE by Eap4 followed by MSA₂PV-AMC turnover by NE. (●) Reaction consists of 17.7 nM CG, 15.4 nM NE, 0 nM Eap4. (▼) Reaction consists of 17.7 nM CG, 0 nM NE, 17.7 nM Eap4. (▲) Reaction consists of 17.7 nM CG, 15.4 nM NE, 17.7 nM Eap4. (■) Reaction consists of 0 nM CG, 15.4 nM NE, 17.7 nM Eap4.

Table 3Kinetic rate constant estimates of neutrophil elastase inhibition determined by progress curve analysis^a

Inhibitor	k_1 ($M^{-1}s^{-1}$)	SD	k_3 (s^{-1})	SD	k_4 ($M^{-1}s^{-1}$)	SD	k_5 (s^{-1})	SD	K_i (nM)	SD
Eap3 WT	2.2×10^5	1.4×10^5	7.0	0.6	4.5×10^4	2.7×10^4	2.1×10^{-4}	2.6×10^{-4}	3.5	2.5
Eap3 R311M	5.1×10^5	2.4×10^5	7.2	0.6	9.0×10^4	1.7×10^3	4.7×10^{-4}	2.8×10^{-4}	5.2	3.0
Eap3 T346A	5.8×10^5	2.5×10^5	6.9	0.3	7.1×10^4	3.8×10^3	3.1×10^{-3}	3.5×10^{-5}	44.7	2.5
Eap3 N348A	2.7×10^6	3.4×10^6	7.0	0.6	5.4×10^4	7.3×10^3	2.0×10^{-3}	1.4×10^{-4}	40.7	6.1
Eap3 T346A/N348A ^b	-	-	8.2	0.2	-	-	-	-	532	24
Eap4 WT	3.7×10^5	1.3×10^5	7.3	0.3	1.4×10^6	1.4×10^5	3.7×10^{-4}	4.0×10^{-5}	0.27	0.02
Eap4 R416M	7.1×10^5	8.4×10^5	5.1	0.3	1.3×10^6	1.7×10^5	2.3×10^{-4}	5.5×10^{-5}	0.18	0.06
Eap4 T451A ^b	-	-	7.3	0.4	-	-	-	-	25.2	2.0
Eap4 P452A	3.1×10^5	2.0×10^5	5.6	0.7	7.1×10^5	1.5×10^5	2.2×10^{-4}	6.9×10^{-5}	0.3	0.2
Eap4 N453A ^b	-	-	5.6	0.3	-	-	-	-	12.7	1.3
Eap4 T451A/P452A	3.6×10^5	1.8×10^5	7.1	0.6	8.6×10^5	1.8×10^5	9.4×10^{-4}	1.2×10^{-4}	1.2	0.4
Eap4 P452A/N453A	2.4×10^5	4.1×10^4	7.6	0.4	5.9×10^5	1.1×10^5	1.4×10^{-3}	9.5×10^{-5}	2.4	0.6
Eap4 T451A/N453A ^b	-	-	7.3	0.2	-	-	-	-	23.7	1.0
Eap4 TPN ^c	3.3×10^5	7.4×10^4	8.0	0.6	6.8×10^5	7.4×10^4	2.8×10^{-3}	1.5×10^{-4}	4.1	0.6
Eap4 P386C-BMP ^d	9.8×10^5	4.7×10^5	4.6	1.2	1.3×10^6	1.8×10^5	3.2×10^{-4}	5.0×10^{-4}	0.21	0.09
Eap4 P452C-BMP ^d	3.1×10^5	3.4×10^4	5.9	0.7	3.6×10^5	1.2×10^5	3.9×10^{-3}	7.7×10^{-4}	11.2	3.5
Eap4 PTN-BMP ^{b,e}	-	-	6.0	0.2	-	-	-	-	141.5	12.9

^a Microscopic rate constant estimates are the average of at least three individual sets of 45 progress curves (three [NE], three [substrate] per [NE], and five [inhibitor] per [substrate]).^b Due to the loss of the time-dependent inhibition characteristic, mutants were analyzed by a classical 5×5 inhibition assay. K_i estimated by globally fitting three assay sets. Error is the standard error of the global fit.^c Eap4 T451A/P452A/N453A.^d The proline to cysteine mutants were crosslinked using the homo-bi-functional crosslinker 1,1-bismaleimidotriethyleneglycol (P386C crosslink oblates the EapH1-binding interface, whereas P452C crosslink oblates the EapH2-binding interface).^e Eap4 P386C/T451A/N453A 1,1-bismaleimidotriethyleneglycol crosslink.

(BMP) to create dimeric inhibitors with only one accessible NE-binding site (*i.e.* Eap4-P386C-BMP for the conserved site and Eap4-P452C-BMP for the alternative site). The Eap4-P386C-BMP protein resulted in WT-like inhibition of NE with an apparent K_i of 0.21 nM. By contrast, the Eap4-P452C-BMP protein displayed inflated inhibition of NE with an apparent K_i of 11.2 nM; it was noteworthy that this ~41-fold inflation was comparable to that observed for the Eap4-T451A and Eap4-453A mutants earlier. Interestingly, when we prepared a crosslinked form of a mutant designed to disrupt both sites simultaneously (*i.e.* Eap4-PTN-BMP), we found that its apparent K_i was inflated ~524-fold to 141.5 nM. The fact that these threonine and asparagine mutations appeared to contribute more to inhibition in the crosslinked protein than in the noncrosslinked protein suggested that the comparatively weaker, alternative site influenced the overall properties we observed for the noncrosslinked Eap4 mutants described in the previous subsection. Together, these data showed that Eap4 can simultaneously inhibit two molecules of NE by way of its tighter-binding conserved site and its relatively weaker-binding alternative site. They also indicated that although Eap4 can bind to and inhibit both CG and NE, it exhibited an overall binding preference for NE similarly to EapH1 (12).

Eap4 ternary complexes with NE form stable dimers in solution

The NE/Eap4/NE asymmetric unit showed two instances of the ternary complex arranged as a dimer lacking cyclic or dihedral symmetry (Figs. 3B and S5A). In this orientation, the C terminus of both Eap4 proteins meet in a “tail-to-tail” fashion. Because Eap4 is the C terminal-most domain within Eap and because the EBI-PISA server suggested that such dimers might be stable structures (22), we wondered if these

dimers of NE/Eap4/NE might be biologically relevant complexes. To explore this possibility further, we used analytical scale size-exclusion chromatography in-line with small angle X-ray scattering (SEC-SAXS) to characterize the solution structure of NE/Eap4/NE. We used CG/EapH2/NE as a control in these experiments because it shared the overall arrangement of the ternary complex (Fig. 1E) but did not show tail-to-tail dimerization in any of the crystal structures available for this complex (17).

We initially collected SEC-SAXS data on CG/EapH2/NE (Fig. 3C and Table 4). The sample behaved as a single, symmetric peak in SEC indicating it lacked aggregation. Guinier analysis revealed an R_g value of 28.3 ± 0.2 Å, while molecular weight estimation from the SAXS profile yielded ~49 kDa consistent with the CG/EapH2/NE assembly seen in various crystal structures. From the maximal dimension (D_{max}) estimation of 94 Å, we generated an *ab initio* density envelope matching the overall arrangement in the crystal structure (Fig. 3, D and G). Most importantly, a theoretical SAXS profile calculated from the CG/EapH2/NE model showed excellent agreement with the experimental SAXS data, as judged by a goodness of fit χ^2 value of 0.86 (Fig. 3, C and D). We then performed an equivalent analysis on NE/Eap4/NE (Fig. 3E and Table 4). Although this sample also behaved as a single peak in SEC, Guinier analysis revealed an R_g value of 36.8 ± 0.3 Å, while molecular weight estimation from the SAXS profile yielded ~111 kDa. From the D_{max} estimate of 110 Å obtained from the pair-distribution function [$P(r)$], we generated an *ab initio* density envelope matching the dimerized NE/Eap4/NE ternary complex (Fig. 3, E and F). Fitting of a theoretical SAXS profile calculated from the dimerized NE/Eap4/NE ternary complex to the experimental data gave an excellent fit, as judged by a χ^2 value of 1.41, whereas fitting a model for the ternary complex alone resulted in a poor χ^2 value of 123.6. The existence of stable dimers for the NE/Eap4/NE ternary

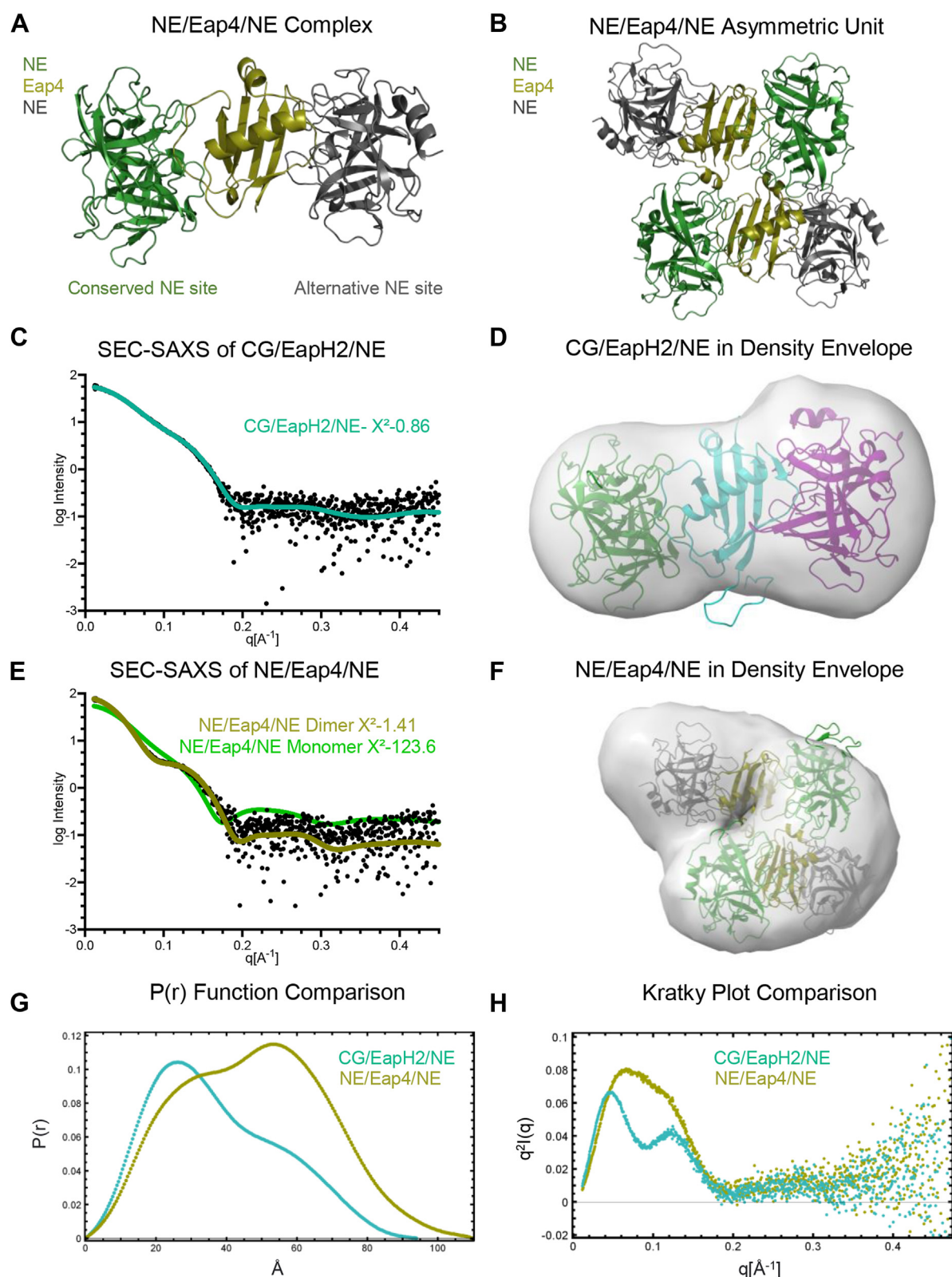


Figure 3. Ternary complexes of Eap4 bound to two copies of NE exist as stable dimers in solution. *A*, cartoon representation of an individual ternary complex of Eap4 (gold) bound to two copies of NE. The NE at the so-called conserved site (dark green) shares a binding mode with the other EAP domains studied here, while the NE at the so-called alternative site (dark gray) shares an NE-binding mode similar to that of EapH1. *B*, cartoon representation of the two NE/Eap4/NE ternary complexes within the asymmetric unit of the crystal. Note that the two copies of the ternary complex are arranged as a "tail-to-tail" dimer lacking cyclic or dihedral symmetry. *C*, SEC-SAXS profile of CG/EapH2/NE (black circles) along with the fit for a model obtained from a structure of the CG/EapH2/NE ternary complex (teal). *D*, a model of the CG/EapH2/NE crystal structure docked into the density envelope calculated from SEC-SAXS data. *E*, SEC-SAXS profile of NE/Eap4/NE (black circles) along with the fits for a model of the dimerized NE/Eap4/NE ternary complex (gold) and a model of the monomeric NE/Eap4/NE ternary complex (light green). *F*, a model of the dimerized NE/Eap4/NE ternary complex docked into the density envelope

Table 4

Overview of structural parameters and apparent molecular weights derived from SAXS experiments

Protein sample ^a	Guinier Rg (Å)	Guinier I(0)	Dmax (Å)	Vc (kDa) ^b	MW Theor (kDa)	Simple Scattering ID	SASBDB ID
CG	18.6 ± 0.1	15.3 ± 0.1	69	20	25	XSDUKX7V	SASDVK2
CG/EapH2/NE	28.3 ± 0.2	56.0 ± 0.2	94	49	62	XSRGKCXV	SASDVL2
NE/Eap4/NE	36.8 ± 0.3	82.0 ± 0.5	110	111	117	XSMG3TCC	SASDVM2
Eap	40.3 ± 0.6	53.1 ± 0.7	153	55	50	XS5RITMZ	SASDVF2
Eap/4CG	50.3 ± 0.9	146 ± 2.3	202	117	152	XSOLKLTA	SASDVG2
Eap/4NE	41.2 ± 1.6	15.2 ± 0.5	106	103	144	XSKTAC2M	SASDVH2
Eap34	24.7 ± 0.3	18.7 ± 0.2	85	20	24	XSWGFWZ	SASDVN2
Eap34/2CG/2NE	53.7 ± 1.1	134 ± 1.2	194	211	243	XSALCU1A	SASDVP2

^a SAXS data for all samples were obtained by SEC-SAXS, except for Eap34/2CG/2NE, which was obtained by HT-SAXS.^b Volume of correlation (Vc) statistic provides an estimate of the molecular weight, presented here in kDa.

complex but not CG/EapH2/NE could also be shown by comparing the P(r) functions and Kratky plots for these two samples (Fig. 3, G and H).

Ternary complexes of multidomain Eap truncations also form stable dimers in solution

Since full-length Eap contains four discrete subdomains capable of binding and inhibiting two NSPs independently (Figs. 1 and 2), we wondered whether the dimerized Eap4 complexes characterized above might also form in larger fragments of Eap that were fully saturated with NSPs. We again explored this possibility using SAXS studies, this time using a combination of SEC-SAXS and more conventional HT-SAXS that examined the behavior of samples across various concentrations. When performing HT-SAXS, samples were carefully prepared to yield desired stoichiometries of the respective components at concentrations high enough to ensure saturation of the binding sites present. We initially planned to study full-length Eap in these experiments. Although some insights were achievable using full-length Eap (Fig. S6 and Table 4), detailed studies on this protein in its fully saturated state were confounded by insufficient solubility of the samples that we could not circumvent using alternative buffer conditions. Thereafter, we used a two-domain fragment of Eap for these studies consisting of domains Eap3 and Eap4 (hereafter Eap34).

We initially characterized Eap34 using SEC-SAXS (Fig. 4, A and B and Table 4). As with full-length Eap (Fig. S6), Eap34 behaved as a single, symmetric peak in SEC, indicating it was devoid of aggregation. Guinier analysis revealed an Rg value of 24.7 ± 0.3 Å, while molecular weight estimation from the SAXS profile yielded ~20 kDa consistent with a monomeric form of Eap34. From the Dmax estimation of 85 Å, we generated an electron density envelope using DENSS (Fig. 4, B and F). We utilized the comparative modeler program in ChimeraX to connect Eap3 and Eap4 with a linker that was treated as flexible during conformational sampling and multistate structural modeling using BILBOMD. The resulting model had a χ^2 of 0.81, demonstrating good agreement with the experimental SAXS data (Fig. 4, A and B).

Thereafter, we conducted HT-SAXS experiments on Eap34 bound to two equivalents of CG and NE at concentrations of 5, 2.5, and 1.25 mg/ml (Fig. 4, C–F and Table 4). As there was no evidence of aggregation or discernible difference in the SAXS profiles, we focused on SAXS data collected on the 5 mg/ml sample (Fig. 4, C and E). The Guinier approximation yielded an Rg of 53.7 ± 1.1 Å, which was larger than that of the dimerized NE/Eap4/NE ternary complex described earlier. We again used the comparative modeler program in ChimeraX to connect the ternary complex structures of CG/Eap3/NE and CG/Eap4/NE (Fig. 1) with a linker and treated it as flexible during conformational sampling in BILBOMD. The resulting model achieved a χ^2 of 0.33 for one state that matched the reconstructed *ab initio* envelope (Fig. 4, D and F). However, including a second state resulted in a significant improvement of the χ^2 to 0.12, as determined by MultiFoXS. This indicated a level of flexibility between the CG/Eap3/NE and CG/Eap4/NE regions. Although Eap34 is a fragment that represents approximately half of the full-length Eap protein, these results strongly suggested that each CG- and NE-binding site was accessible in multidomain forms of Eap. Moreover, these observations support the proposal that NSP binding-induced dimerization of Eap4 was a feature retained in progressively larger forms of the Eap protein. Given these complexes' unusual nature and overall architecture (Fig. 4, G and H), we propose that higher-order structures such as these may display unexpected functional properties.

Discussion

Whereas serine protease inhibitors have been known for nearly 100 years (23), the identification of EAP domains as NSP inhibitors was made relatively recently (11). Since there was little information available at that time, the existence of three unique EAP family members presented a challenge because initial work suggested that all three EAP proteins had essentially similar functional properties (11). EapH1 emerged into our paradigm for understanding the structure/function relationships of this family, primarily due to the NE/EapH1 crystal structure that provided the first structural insights into EAP function as NSP inhibitors. Using more rigorous assay

calculated from SEC-SAXS data. G, comparison of the pair-distribution (P(r)) functions for CG/EapH2/NE (teal) and NE/Eap4/NE (gold). Both the shape of the individual P(r) curves as well as the Dmax indicate a difference in the solution arrangements for these two complexes. H, comparison of the Kratky plot for CG/EapH2/NE (teal) and NE/Eap4/NE (gold). As with the P(r) curves, the differences in the profile of these plots belie a difference in the solution structures of these two complexes.

methods, we discovered that EapH1 acts as a potent, time-dependent inhibitor of both NE (13) and CG (12), but with K_i values that differ by ~460-fold between these two NSPs (c.f. $K_{i, NE} = 21$ pM and $K_{i, CG} = 9.8$ nM). Our subsequent crystal structure of CG/EapH1 provided a basis for understanding these functional differences and showed that EapH1 uses a similar binding mode to bind both CG and NE (12). This observation, as well as the lack of any conclusive insights available through sequence analysis (Fig. S1), led to the hypothesis that EapH2, as well as the individual domains within Eap, would behave similarly to EapH1. However, our initial work on EapH2 suggested this was likely not the case (15, 16). Through a combination of solution NMR and functional assays, we made the surprising observation that EapH2 can simultaneously bind and inhibit CG and NE (17), while our crystal structure of the CG/EapH2/NE complex defined the binding sites of each NSP within EapH2 (17). Although initial structure/function characterization of CG and NE inhibition by Eap1 and Eap2 suggested these domains of Eap might behave similarly to EapH2 (18), here we showed that both Eap1 and Eap2 form ternary complexes with CG and NE that block both NSPs simultaneously (Figs. 1, 2, and S2). We also found that both Eap3 and Eap4 form analogous ternary complexes to CG/EapH2/NE (Figs. 1, 2, and S2). Even though Eap1, Eap2, Eap3, and Eap4 exist as discrete subdomains within the modular Eap fusion protein, their overall structure/function properties bear far greater similarities to EapH2 than EapH1 insofar as NSP inhibition is concerned. Therefore, we conclude that EapH2 is a more appropriate paradigm for understanding and describing NSP inhibition by EAP domains than EapH1.

A major outcome of our earlier efforts and the work we present here is that extensive data are now available for EapH1, EapH2, and the individual domains of full-length Eap. This allows us to draw several broad conclusions about the structure/function properties of EAP domains. First, all of the EAP domains aside from EapH1 can form ternary complexes, thereby inhibiting CG and NE simultaneously ((17); Figs. 1, 2, and S2). This was described in detail in the preceding paragraph, but it bears repeating here because similar properties have not been assigned to any other family of serine protease inhibitors. Indeed, this unusual attribute is the rule among EAP domains rather than the exception. Second, all WT EAP domains examined thus far exhibit time-dependent inhibition of CG and NE ((12, 13, 16–18); Tables 2 and 3), with the lone exception being CG inhibition by Eap4. Time-dependent inhibition is distinguished by a change in reaction velocity *versus* time, ultimately leading to a steady-state rate (24). Such inhibitors are typically characterized by slow off-rates from their respective enzymes. EAP domains certainly meet this criterion, exhibiting dissociation rate constants ranging from 10^{-5} to 10^{-4} s $^{-1}$ for NSPs. Moreover, loss of residues at the NSP/EAP interface that diminish the apparent K_i almost always increases the dissociation rate and, in several cases, results in loss of this time-dependent inhibitory characteristic. Finally, although all EAP domains inhibit CG and NE, detailed kinetic studies have shown that these molecules are generally ~10-fold weaker

inhibitors of CG when compared to NE ((12, 13, 16–18); Tables 2 and 3). This feature arises from an increase in the association rate of the EAP domain inhibitor for NE when compared to CG. While the reasons for this disparity are unclear, it does not seem to be due to increased shape complementarities or buried surface areas for NE/EAP complexes compared to CG/EAP complexes (Table S1). Therefore, this could be an attractive topic for future investigation.

While there is value in identifying the shared structure/function features of EAP domains, it is equally important to consider the outliers and the biological ramifications thereof. In this regard, Eap4 stands out for multiple reasons. First, we found that Eap4 is the weakest inhibitor of CG ($K_i = 434$ nM) among all EAP domains examined thus far ((12, 17, 18); Table 2). This comparatively low affinity does not prevent Eap4 from forming a CG/Eap4/NE ternary complex or inhibiting both NSPs simultaneously (Figs. 1, 2, and S2). Second, we observed unusual kinetic parameters for NE inhibition by Eap4 that led us to discover that this EAP domain uniquely contains two relatively high-affinity NE-binding sites (Figs. 3, S3–S5, and Table 3). Whereas the first of these corresponds to the conserved NE-binding site shared by all EAP domains except EapH1 and is exceptionally high affinity ($K_i = 0.21$ nM), the second represents an alternative NE-binding site that is structurally equivalent to its CG-binding site. This so-called alternative site bears a strong resemblance to that of EapH1, wherein a single binding site accommodates both CG and NE albeit at the cost of binding one NSP preferentially to the other (11–13). In this regard, the ~40-fold difference in affinity of this alternative site for NE compared to CG suggests that NE binding at this site will predominate in the absence of other factors. Finally, we found that dimers of the NE/Eap4/NE ternary complex present in the crystallographic asymmetric unit are stable structures in solution (Figs. 3, S5, and Table 4). These dimers form in a tail-to-tail fashion, which is only possible since Eap4 represents the naturally occurring C-terminus of the Eap protein. These dimers also form only following NSP binding since Eap4 on its own, the two-domain fragment Eap34, and full-length Eap itself are all monomers (Figs. 4, S6, and Table 4). Perhaps most significantly, these dimers appear compatible with NSP binding at all available sites in multidomain Eap fragments, as judged by SAXS studies of Eap34 in the presence of saturating levels of CG and NE (Fig. 4).

Considering the unique biological contexts where Eap might function leads us to one conclusion and one prediction. Principally, we conclude that Eap is a polyvalent inhibitor of NSPs (Figs. 1, 2, and Tables 1–3). Although Eap is produced at a basal level by *S. aureus*, its expression levels increase when the bacterium is exposed to neutrophil granule components (25). This is most relevant when *S. aureus* finds itself in the neutrophil phagosome, where the concentrations of NSPs are believed to approach ~1 mM (26, 27). Even if these estimates are in error by 100-fold or more, the NSP concentration within the phagosome would still be far above the observed K_i for any of the individual NSP-binding sites on Eap ((18); Tables 2 and 3). In this scenario, all available NSP-binding sites on Eap secreted

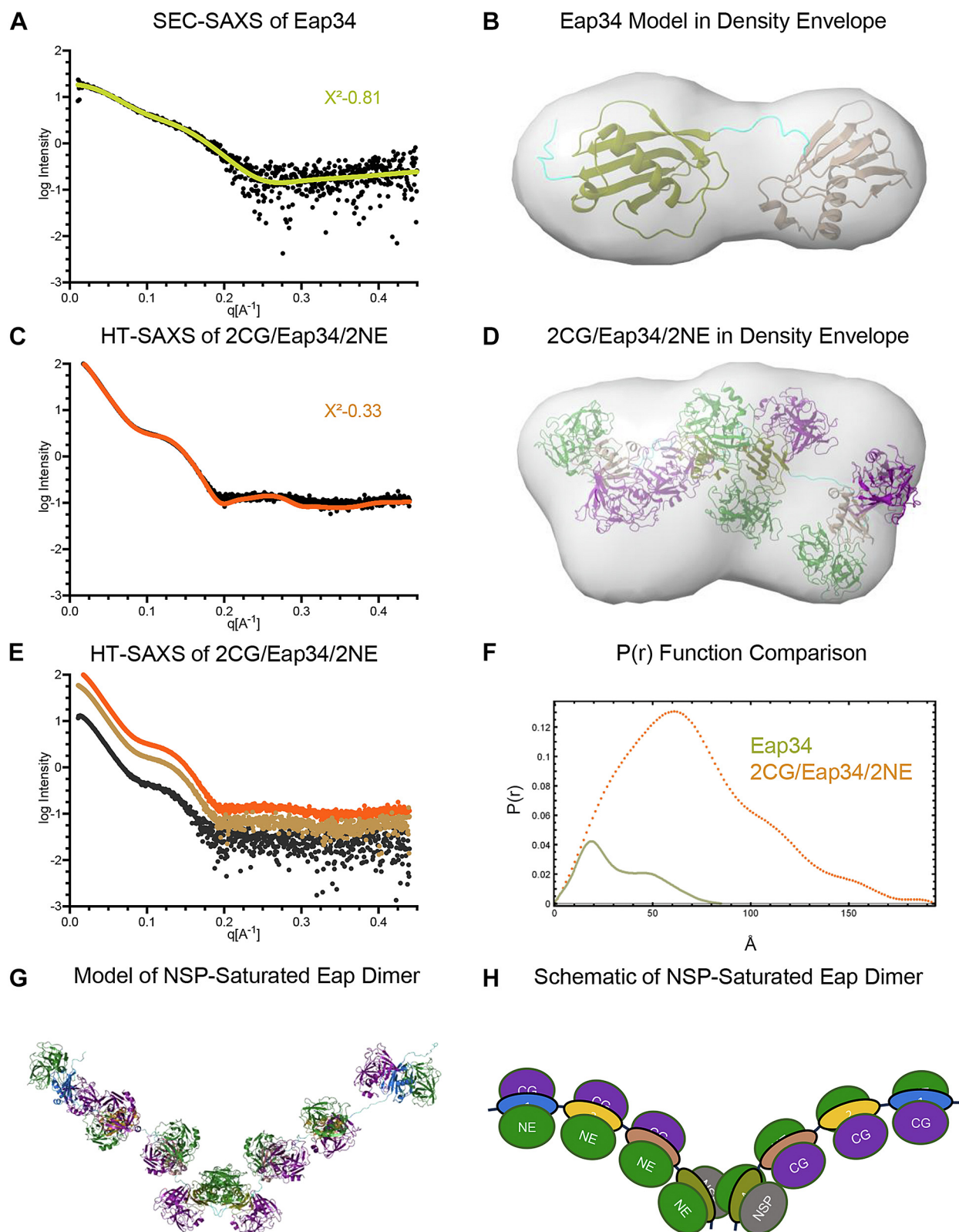


Figure 4. NSP binding-induced dimers can also form in larger fragments of full-length Eap. *A*, SEC-SAXS profile of Eap34 (black circles) along with the fit for a model obtained by linking empirical structures of Eap3 and Eap4 with a flexible linker (dark yellow). *B*, a single-state model of the Eap34 structure docked into the density envelope calculated from SEC-SAXS data. Note that Eap3 (bronze) and Eap4 (gold) are colored identically to Figure 1, while the flexible linkers present in protein are colored in teal. *C*, HT-SAXS profile of Eap34 bound to two equivalents of CG and NE (i.e. 2CG/Eap34/2NE) obtained at 5 mg/ml concentration (black circles). The fit for a model obtained by extending a dimer of the CG/Eap4/NE ternary complex N-terminally with CG/Eap3/NE connected via a flexible linker is also shown (orange). *D*, a single-state model of dimerized 2CG/Eap34/2NE docked into the density envelope calculated from HT-SAXS data. *E*, comparison of the HT-SAXS profiles obtained at 5 mg/ml (orange), 2.5 mg/ml (gold), and 1.0 mg/ml (black) total protein for samples of 2CG/Eap34/2NE. The consistent profiles obtained at these three concentrations strongly suggest that these samples were free of concentration-dependent

into the phagosome should be saturated with ternary complexes like those described here (Fig. 1H). We attempted to characterize full-length Eap saturated with NSPs through SAXS methods, as mentioned earlier; unfortunately, these samples were not sufficiently soluble to proceed further. Nevertheless, preliminary characterizations of full-length Eap in the presence of either CG or NE were consistent with each NSP binding at their respective sites (Fig. S6). A two-domain Eap fragment (*i.e.* Eap34) also showed evidence of full NSP binding in SAXS studies. In the future, pursuing structural analysis of NSP-saturated Eap using various electron microscopy methods may ultimately prove more productive.

Secondarily, we propose that dimerization of NSP-bound Eap may have important biological ramifications. Ternary complexes of Eap4 seem uniquely capable of forming stable dimers in solution (Figs. 3 and 4). We think this is more than a biochemical coincidence, as such structures could form in full-length Eap provided that sufficient NSPs are present. At first pass, it is difficult to envisage how forming higher-order structures like these might influence the ability of Eap to inhibit NSPs. Considering that NSP-bound Eap (i) would be highly extended in shape, (ii) would present a repeating array of unique binding surfaces by virtue of the bound NSPs decorating its individual domains, and (iii) would have remarkable cationic character due to the overwhelming basic nature of the components (pI, Eap = 10.0; pI, CG = 11.5, pI, NE = 10.6) leads to an interesting scenario (Fig. 4, G and H). Eap has previously been reported to be a DNA-binding protein that can block the formation of neutrophil extracellular traps (NETs), as they are commonly known (28). However, the molecular mechanisms underlying this activity remain unexplored. We propose that the NSP/Eap complexes uncovered here could contribute to this process and that the dimerization of these complexes may play a significant role. Separately, since NSPs are well-established components of NETs (29, 30), we suspect that these enzymes bound to extracellular DNA could recruit Eap to pre-existing NETs and alter NET function. Exploring these possibilities further will remain a priority for future investigations.

Experimental procedures

Protein samples

Human NE was procured from Elastin Products Company, while human neutrophil CG was obtained from Innovative Research. Derivatives of the prokaryotic expression vector pT7HMT encoding either WT or mutant forms of EapH1, EapH2, Eap1, Eap2, Eap3, Eap4, Eap, and Eap34 were obtained from GenScript USA Inc (13, 16, 18, 31). Both WT and mutant EAP domains were expressed in *Escherichia coli* BL21(DE3) cells and purified as previously described (13, 16). The concentrations of WT and mutant EAP domains were assessed

spectrophotometrically using a DS-11 Spectrophotometer from DeNovix and the following extinction coefficients, which were derived from the protein compositions using ExPASy ProtParam: Eap1, $\epsilon_{280} = 7.45 \text{ mM}^{-1} \text{ cm}^{-1}$; Eap2, $\epsilon_{280} = 5.96 \text{ mM}^{-1} \text{ cm}^{-1}$; Eap3, $\epsilon_{280} = 15.93 \text{ mM}^{-1} \text{ cm}^{-1}$; Eap4, $\epsilon_{280} = 8.94 \text{ mM}^{-1} \text{ cm}^{-1}$; Eap, $\epsilon_{280} = 41.26 \text{ mM}^{-1} \text{ cm}^{-1}$; Eap34, $\epsilon_{280} = 26.36 \text{ mM}^{-1} \text{ cm}^{-1}$.

Protein crystallography and structural analysis

Crystallization samples of protein complexes were prepared by mixing the appropriate molar ratios of the required monomers. Ternary complexes of Eap1 and Eap2 were exchanged into a buffer of 10 mM Hepes (pH: 7.4), 50 mM NaCl and concentrated to 10 mg/ml. For other complexes, samples were exchanged into a buffer of 10 mM Tris (pH: 7.4), 50 mM NaCl and concentrated to 5 mg/ml. The concentration of each sample was determined based on the calculated extinction coefficient of the respective complexes. Initial crystallization conditions were determined by vapor diffusion of sitting drops using a Crystal Gryphon liquid handling system (Art Robbins Instruments, LLC) and the Crystal Screen HT, Index HT, and PEGRx HT screening matrices (Hampton Research Corp). Screening droplets consisted of a total volume of 0.4 μl and were prepared as both 3:1 and 2:2 ratios of sample to precipitant solution prior to equilibration alongside 100 μl of precipitant solution. Optimization of individual crystallization conditions was carried out through vapor diffusion of hanging drops and involved systematic adjustments to sample concentration, buffer pH, precipitant concentration, and potential utilization of microseeding techniques. All initial crystallization screens and optimization studies were conducted at a temperature of 20 °C. Further details regarding the finalized crystallization conditions and the composition of individual models are elaborated in paragraphs at the end of this subsection.

Samples of protein crystals were harvested, incubated briefly in cryopreservation buffer, and flash cooled in liquid N₂ prior to collecting monochromatic X-ray diffraction data using beamlines 22-BM and 22-ID of the Advanced Photon Source of Argonne National Laboratory. Diffraction data were indexed, integrated, and scaled using a combination of the HKL2000 (32) and XDS (33) software packages. Structures were solved by molecular replacement using PHASER (34) as implemented within the PHENIX suite (35) using specific search models appropriate to each target structure. Initial solutions were improved through automated rebuilding using PHENIX.AUTOBUILD (35) and manual modifications using COOT (36). Crystallographic models were refined through a combination of protocols including simulating annealing, reciprocal space positional refinement, B-factor refinement, and TLS refinement using PHENIX.REFINE (35). Each refined

effects and/or aggregation. *F*, comparison of the pair-distribution ($P(r)$) functions for Eap34 (gold) and 2CG/Eap34/2NE (orange). Both the shape of the individual $P(r)$ curves as well as the D_{max} indicate a difference in the solution arrangements for these samples of Eap34. *G*, putative model of an extended structure formed upon dimerization of full-length Eap in the presence of saturating levels of CG and NE. All proteins are shown as ribbon diagrams and colored identically to Figure 1. *H*, schematic representation of the structure shown in panel G, where the NSP-binding site of Eap4 that can be occupied by either NE or CG is colored gray.

model and the accompanying structure factors have been deposited in the PDB under the accession codes provided elsewhere in the text (Table 1). Detailed information regarding diffraction data collection statistics and model properties for each structure is also available in the same location (Table 1).

Crystals of the ternary complex of Eap1 bound to CG and NE were obtained from a hanging drop that had been prepared by mixing 1 μ l of 5 mg/ml protein with 1 μ l of a precipitant solution consisting of 0.1 M Bis-Tris (pH: 6.9), 0.2 M sodium potassium tartrate tetrahydrate, 14% (w/v) PEG-8000. Individual crystals were cryoprotected prior to harvest by briefly soaking in a buffer of precipitant solution supplemented with 11% (v/v) glycerol. Monochromatic X-ray diffraction data extending to 3.63 Å limiting resolution were collected using APS beamline 22-ID and were indexed, integrated, and scaled in the space group $P2_12_12_1$ using XDS. The structure was solved by molecular replacement using the structures of Eap1 (PDB entry 8D4O), CG (PDB entry 1AU8), and NE (PDB entry 1HNE) as search models and contains six ternary complexes in the asymmetric unit. The final model consists of six complete copies of Eap1, six complete copies of CG, and six complete copies of NE. Using the complex defined by chains A, B, and C as a basis for comparison, the remaining complexes superimpose with RMSD values of 0.282 Å (chains D, E, and F), 0.336 Å (chains G, H, and I), 0.512 Å (chains J, K, and L), 0.408 Å (chains M, N, and O), and 0.584 Å (chains P, Q, and R), respectively. The final model and refined structure factors have been deposited in the PDB as entry 8D7L.

Crystals of the ternary complex of Eap2 bound to CG and NE were obtained from a hanging drop prepared by mixing 1 μ l of 5 mg/ml protein with 1 μ l of a precipitant solution consisting of 0.1 M citric acid (pH: 3.6), 0.15 M lithium sulfate monohydrate, 12% (w/v) PEG-6000. Individual crystals were cryoprotected prior to harvest by briefly soaking in a buffer of precipitant solution supplemented with 23% (w/v) PEG-6000. Monochromatic X-ray diffraction data extending to 3.10 Å limiting resolution were collected using APS beamline 22-ID and were indexed, integrated, and scaled in the space group $P2_1$ using HKL2000. The structure was solved by molecular replacement using the structures of Eap2 (PDB entry 1YN3), CG (PDB entry 1AU8), and NE (PDB entry 1HNE) as search models and contains four ternary complexes in the ASU. The final model consists of four complete copies of Eap2, four complete copies of CG, and four complete copies of NE. Using the complex defined by chains A, B, and C as a basis for comparison, the remaining complexes superimpose with RMSD values of 0.269 Å (chains D, E, and F), 0.625 Å (chains G, H, and I), and 0.367 Å (chains J, K, and L), respectively. The final model and refined structure factors have been deposited in the PDB as entry 8D7K.

Crystals of the ternary complex of Eap3 bound to CG and NE were grown using a hanging drop method, where 3 μ l of 5 mg/ml protein was mixed with 1 μ l of a precipitant solution containing 0.1 M imidazole (pH: 6.8), 0.2 M ammonium citrate, 21% (w/v) peg-2K-MME. Prior to harvesting, individual crystals were briefly soaked in buffer of precipitant solution supplemented with 14% (v/v) glycerol for cryoprotection. X-

ray diffraction data extending to 1.95 Å resolution were collected at APS beamline 22-ID and were indexed, integrated, and scaled in space group $P2_12_12_1$ using HKL2000. The structure was solved by molecular replacement using the structures of Eap2 (PDB: 1YN3), CG (PDB: 1AU8), and NE (PDB: 1HNE) as search models and contains a single ternary complex in the ASU. The final model consists of a complete copy of Eap3, CG, and NE, associated N-linked glycan chains, and 205 ordered solvent molecules. The final refined model and structure factors have been deposited in the PDB as entry 9ASX.

Crystals of the binary complex of Eap4 with NE were obtained *via* the sitting drop method, where 5 mg/ml protein was mixed with a precipitant solution consisting of 0.1 M MES (pH: 6.5), 10 mM zinc sulfate heptahydrate, 25% (v/v) peg-550MME in a 3:1 ratio. Individual crystals were cryoprotected by briefly soaking in a buffer of precipitant solution supplemented with 10% (v/v) peg-400. X-ray diffraction data extending to 1.75 Å resolution limit were collected at APS beamline 22-ID and were indexed, integrated, and scaled in space group $P222$ using HKL2000. The structure was solved by molecular replacement using the structures of Eap2 (PDB entry 1YN3) and NE (PDB entry 1HNE) as search models and contains one binary complex in the ASU. The final model consists of complete copies of Eap4 and NE, associated N-linked glycan chains, an ordered sulfate ion, and 246 ordered solvent molecules. The refined model and structure factors have been deposited in the PDB as entry 9ASS.

Crystals of the Eap4 ternary complex bound to CG and NE were obtained *via* the hanging drop method, where 3 μ l of 5 mg/ml protein were mixed with 1 μ l of a precipitant solution containing 0.1 M sodium citrate (pH: 5.5), 12% (w/v) PEG-6K. Individual crystals were cryoprotected by briefly soaking in a buffer of precipitant solution supplemented with 23% (v/v) 2,4-methylpentanediol. X-ray diffraction data extending to 2.10 Å limiting resolution were collected at APS beamline 22-ID and were indexed, integrated, and scaled in space group $P1$ using HKL2000. The structure was solved by molecular replacement using the structure of Eap4 bound to NE (PDB entry 9ASS) and CG (PDB entry 1AU8) as search models and contains four ternary complexes in the ASU. The final model consists of four complete copies of Eap4, four complete copies of CG, four complete copies of NE, associated N-linked glycan chains, and 770 ordered solvent molecules. Using the complex defined by chains A, B, and C as a basis for comparison, the remaining complexes superimpose with RMSD values of 0.269 Å (chains D, E, and F), 0.625 Å (chains G, H, and I), and 0.367 Å (chains J, K, and L), respectively. The refined model and structure factors have been deposited in the PDB as entry 9ATK.

Crystals of the Eap4 ternary complex bound to two molecules of NE were obtained *via* the hanging drop method, where 3 μ l of 5 mg/ml protein were mixed with 1 μ l of a precipitant solution containing 0.1 M Tris-Cl (pH: 8.0), 0.2 M ammonium acetate, 17% (w/v) PEG-10K. Individual crystals were cryoprotected by briefly soaking in a buffer of precipitant solution supplemented with 18% (v/v) glycerol. X-ray diffraction data extending to 2.05 Å resolution were collected at APS

beamline 22-ID and indexed, integrated, and scaled in space group $P2_12_12_1$ using HKL2000. The structure was solved by molecular replacement using the structure of Eap4 bound to NE (PDB entry 9ASS) and NE (PDB entry 1HNE) as search models and contains two ternary complexes in the ASU. The final model consists of two complete copies of Eap4, four complete copies of NE, associated N-linked glycan chains, and 338 ordered solvent molecules. The two complexes superimpose with an RMSD value of 0.269. The refined model and structure factors have been deposited in the PDB as entry 9ATU.

Analyses of protein structures

Representations of protein structures were generated from the respective PDB files utilizing PyMol (Schrodinger, LLC). RMSD of superposition for individual proteins or protein–protein complexes was calculated using PyMol. Buried surface area at interfaces, along with automated identification of hydrogen bonds and salt bridges, were determined using EBI-PISA (22). Surface complementarity between interacting proteins was assessed using the program s.c. (37) within the CCP4 suite (38). Protein sequence alignments and evaluations of sequence similarity/identity were performed using Clustal Omega (39).

Preparation of CG, NE, and substrates for assay

Lyophilized human NE (1 mg) was dissolved in 200 μ l of 20 mM sodium acetate/50% (v/v) glycerol (pH: 5.0) and stored at -20°C in 10 μ l aliquots. Enzyme concentration was determined using the deduced protein composition ($\epsilon_{280} = 20.23 \text{ mM}^{-1} \text{ cm}^{-1}$). The NE substrate, methoxysuccinyl-Ala-Ala-Pro-Val-p-nitroanilide (MSA₂PV-pNA) was dissolved in 100% (v/v) DMSO, diluted in 50 mM Hepes/140 mM NaCl/0.05% (v/v) Tween-20 (pH: 7.4) to achieve the working concentration, and quantified using end-point assays ($\epsilon_{280} = 12.3 \text{ mM}^{-1} \text{ cm}^{-1}$). Human CG was stored at 4°C in 50 mM sodium acetate/800 mM NaCl (pH 5.5) and diluted prior to use with 20 mM Hepes/140 mM NaCl/0.05% (v/v) Tween-20 (pH: 7.4). SA₂PF-pNA was dissolved in 100% (v/v) DMSO. The fluorescent substrate, MSA₂PV-AMC was dissolved in 100% (v/v) DMSO to a final concentration of 15.9 mM. All substrates were stored with dessicant at 4°C if dissolved or at -20°C in powdered form.

Simultaneous inhibition of CG and NE by individual Eap domains

To probe the ability of the EAP domains to inhibit both NSPs simultaneously, parallel reaction pairs were prepared consisting of both proteases with their respective substrates with or without an EAP domain inhibitor. To investigate simultaneous inhibition by Eap1, four separate reaction pairs were prepared. Reaction pair one consisted of 15 nM CG incubated with 340 μ M MSA₂PV-AMC in the top well and 15.2 nM NE incubated with 2.35 mM SA₂PF-pNA in the lower well. Reaction pair two consisted of 15 nM CG incubated with 340 μ M MSA₂PV-AMC in the top well with 18.8 nM Eap1

incubated with 2.35 mM SA₂PF-pNA in the lower well. Reaction pair three consisted of 15 nM CG incubated with 340 μ M MSA₂PV-AMC in the top well with 15.2 nM NE incubated with 2.35 mM SA₂PF-pNA and 18.8 nM Eap1 in the lower well. Reaction pair four consisted of 340 μ M MSA₂PV-AMC in the top well with 15.2 nM NE incubated with 2.35 mM SA₂PF-pNA and 18.8 nM Eap1 in the lower well. The reaction components were allowed to incubate for 30 min prior to mixing. The reactions were initiated by mixing 170 μ l of the top well into 170 μ l of the lower well. The final concentrations of the reaction components were 7.5 nM CG, 7.6 nM NE, 9.4 nM Eap1, 170 μ M MSA₂PV-AMC, and 1.175 mM SA₂PF-pNA.

To investigate simultaneous inhibition by Eap2, four separate pairs of reactions were also prepared. Reaction pair one consisted of 16.4 nM CG with 340 μ M MSA₂PV-AMC in the top well and 14.6 nM NE with 2.35 mM SA₂PF-pNA in the lower well. Reaction pair two consisted of 16.4 nM CG with 340 μ M MSA₂PV-AMC in the top well and 20.6 nM Eap2 with 2.35 mM SA₂PF-pNA in the lower well. Reaction pair three consisted of 16.4 nM CG with 340 μ M MSA₂PV-AMC in the top well and 14.6 nM NE with 2.35 mM SA₂PF-pNA and 20.6 nM Eap2 in the lower well. Reaction pair four consisted of 340 μ M MSA₂PV-AMC in the top well and 14.6 nM NE with 2.35 mM SA₂PF-pNA and 20.6 nM Eap2 in the lower well. The reaction components were allowed to incubate for 30 min prior to mixing. The reactions were initiated by mixing 170 μ l of the top well into 170 μ l of the lower well. The final concentrations of the reaction components were 8.2 nM CG, 7.3 nM NE, 10.3 nM Eap2, 170 μ M MSA₂PV-AMC, and 1.175 mM SA₂PF-pNA.

To investigate simultaneous inhibition by Eap3, four separate pairs of reactions were also prepared. Reaction pair one consisted of 35.4 nM CG with 341 μ M MSA₂PV-AMC in the top well and 30.8 nM NE with 2.2 mM SA₂PF-pNA in the lower well. Reaction pair two consisted of 35.4 nM CG with 341 μ M MSA₂PV-AMC in the top well and 77.2 nM Eap3 with 2.2 mM SA₂PF-pNA in the lower well. Reaction pair three consisted of 35.4 nM CG with 341 μ M MSA₂PV-AMC in the top well and 30.8 nM NE with 2.2 mM SA₂PF-pNA and 77.2 nM Eap3 in the lower well. Reaction pair four consisted of 341 μ M MSA₂PV-AMC in the top well and 30.8 nM NE with 2.2 mM SA₂PF-pNA and 77.2 nM Eap3 in the lower well. The reaction components were allowed to incubate for 30 min prior to mixing. The reactions were initiated by mixing 170 μ l of the top well into 170 μ l of the lower well. The final concentrations of the reaction components were 17.7 nM CG, 15.4 nM NE, 38.6 nM Eap3, 170.5 μ M MSA₂PV-AMC, and 1.1 mM SA₂PF-pNA.

To investigate simultaneous inhibition by Eap4, four separate pairs of reactions were also prepared. Reaction pair one consisted of 35.4 nM CG with 341 μ M MSA₂PV-AMC in the top well and 30.8 nM NE with 2.2 mM SA₂PF-pNA in the lower well. Reaction pair two consisted of 35.4 nM CG with 341 μ M MSA₂PV-AMC in the top well and 35.4 nM Eap4 with 2.2 mM SA₂PF-pNA in the lower well. Reaction pair three

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consisted of 35.4 nM CG with 341 μ M MSA₂PV-AMC in the top well and 30.8 nM NE with 2.2 mM SA₂PF-pNA and 35.4 nM Eap4 in the lower well. Reaction pair four consisted of 341 μ M MSA₂PV-AMC in the top well and 30.8 nM NE with 2.2 mM SA₂PF-pNA and 35.4 nM Eap4 in the lower well. The reaction components were allowed to incubate for 30 min prior to mixing. The reactions were initiated by mixing 170 μ l of the top well into 170 μ l of the lower well. The final concentrations of the reaction components were 17.7 nM CG, 15.4 nM NE, 17.7 nM Eap4, 170.5 μ M MSA₂PV-AMC, and 1.1 mM SA₂PF-pNA.

The CG reaction was followed spectrophotometrically (400 nm) while the NE reaction was followed fluorometrically (λ_{ex} : 355 nm, λ_{em} : 455 nm). The buffer consisted of 50 mM Hepes/140 mM NaCl/0.05% (v/v) tween-20/5% (v/v) DMSO (pH: 7.4).

Steady-state kinetic analysis of CG and NE inhibition by Eap3 and Eap4

Methods for measuring inhibition of CG and NE by steady-state kinetics have been previously described (12, 13, 16). Progress curve analysis was used to examine a series of replicate enzyme-catalyzed reactions followed for 30 min. Thirty progress curves were collected for CG (two CG concentrations, three substrate concentrations per CG concentration, and five inhibitor concentrations per substrate concentration), while 45 progress curves were collected for NE (three NE concentrations, three substrate concentrations per NE concentration, and five inhibitor concentrations per substrate concentration). All data were fit to a rapid-equilibrium competitive inhibition mechanism using KinTek Global Kinetic Explorer (40, 41) to derive the estimates of microscopic rate constants. The substrate off-rate (k_2) was linked to the on-rate (k_1) by the estimated Michaelis constant (K_M). The apparent inhibition constant (*i.e.* K_i) was calculated from the experimentally determined on- and off-rate estimates. The values reported are the average of at least three different sets of data, with the SD in each parameter representing the reported errors.

Crosslinking of Eap4-P386C and Eap4-P452C mutants

Protein samples were overexpressed and purified using the general procedures given in an earlier subsection. Prior to crosslinking, the protein was initially buffer exchanged into 50 mM Hepes/500 mM NaCl (pH: 7.0). Tris(2-carboxyethyl) phosphine was then added to the purified protein to achieve a $\sim$ 1.9-fold molar excess to protein concentration and the sample was incubated for 30 min at room temperature. The bifunctional crosslinker BMP was added to the reduced protein to achieve an $\sim$ 8-fold molar excess to protein concentration. The reaction was incubated overnight at room temperature. Following quenching of the reaction by the addition of 2 mM tris(2-carboxyethyl)phosphine, the cross-linked Eap4 protein was further purified by size-exclusion

chromatography over a Sephacryl S-100 HR High Prep 26/60 column that had been previously equilibrated in HBS (pH: 7.4).

SAXS experiments

All SAXS experiments were performed at the SIBYLS beamline 12.3.1 of the Advanced Light Source at Berkeley National Laboratory. This beamline features inline instrumentation and detectors connected to a size-exclusion column (42, 43). Samples of protein complexes were prepared at 5 mg/ml concentration and were separated by a Shodex 803 SEC column that had been connected to an Agilent 1260 Infinity HPLC system operating at a flow rate of 0.65 ml min⁻¹. For EapH2, CG, Eap, CG/EapH2/NE, NE/Eap4/NE, and Eap/4CG, a buffer of 20 mM Hepes (pH: 7.4), 140 mM NaCl was used as the mobile phase; for Eap/4NE, Eap34, and 2CG/Eap34/2NE, a buffer of 20 mM acetate (pH: 4.0), 100 mM L-arginine, 100 mM L-glutamic acid, 300 mM NaCl, 1% (v/v) glycerol was used as the mobile phase. The chromatography system was connected to a flow cell to allow for SAXS measurements. SAXS measurements involved continuous 2 s X-ray exposures collected over a 25 min elution period. Experimental frames were analyzed after subtracting a background frame collected on the mobile phase alone prior to peak detection. The X-ray wavelength was 1.127 Å, while the sample-to-detector distance was 2100 mm. This resulted in scattering vectors (q) ranging from 0.01 Å⁻¹ to 0.45 Å⁻¹.

The Guinier approximation ($I(q) = I(0) \exp(-q^2 R_g^2/3)$, with $qR_g < 1.3$) was used to calculate the radius of gyration (R_g) for each subtracted frame. The elution peak was assessed by comparing the integral ratios to background and R_g relative to the recorded frame using RAW (44). Uniform R_g values across the elution peak indicate a homogeneous sample. Merged SAXS profiles, obtained by integrating multiple frames at the elution peak, were further analyzed. The Guinier plot provided insights into the aggregation state, while the volume of correlation (V_c) estimated the molecular weight (45). The pair distribution function ($P(r)$) was used to calculate the maximal dimension, D_{max} (46).

HT-SAXS was also performed on samples of 2CG/Eap34/2NE. Samples of this complex were prepared at 5, 2.5, and 1.25 mg/ml protein in a buffer of 20 mM Hepes (pH: 7.4), 140 mM NaCl. The X-ray wavelength was 1.127 Å, while the sample-to-detector distance was again fixed at 2100 mm. Experimental frames were analyzed after subtracting a background frame collected on the buffer alone. RAW (44), ATSAS (47), and Scatter were used for data processing as described above.

Processed scattering data from either SEC-SAXS or HT-SAXS experiments were used as the basis for modeling solution ensembles of the respective complex. Crystal structures of ternary complexes for Eap domains 1, 2, 3, and 4 in the presence of CG and NE served as inputs (48). To create a unified structure, individual EAP domains were linked using the comparative modeler program in ChimeraX (49). Multi-state modeling with SAXS profiles was then conducted using

BILBOMD (50) and MultiFoXS (51), wherein all polypeptide chains were connected as rigid bodies. Ten thousand conformations were generated with their corresponding SAXS profiles calculated by FoXS (52). Conformations with the lowest χ^2 values were selected for final assessment and validation.

Data availability

The atomic coordinates and structure factors for crystal structure determinations have been deposited in the Protein Data Bank (<http://wwpdb.org/>) and are publicly available through the accession codes provided in Table 1. Small angle X-ray scattering data, models, and fits have been deposited in the Small Angle Scattering Biological Data Bank (<http://www.sasbdb.org/>) and are publicly available through the accession codes provided in Table 4. SAXS data and models have also been deposited in the Simple Scattering open data repository (<https://simplescattering.com>) using the accession codes provided in Table 4. All other experimental data are available upon request by contacting the corresponding author B. V. G. (geisbrechtb@ksu.edu).

Supporting information—This article contains supporting information.

Acknowledgments—The authors thank Samuel Bouyain, Saurav Misra, and Michael Kanost for helpful discussions throughout the course of this work and for their comments on an initial version of this manuscript. X-ray diffraction data were collected at Southeast Regional Collaborative Access Team (SER-CAT) beamlines 22-BM and 22-ID at the Advanced Photon Source, Argonne National Laboratory. A list of supporting institutions may be found on the SER-CAT website. Use of the Advanced Photon Source was supported by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences, under Contract No. W-31-109-Eng-38. Portions of this work were conducted at the Advanced Light Source (ALS), a national user facility operated by Lawrence Berkeley National Laboratory on behalf of the Department of Energy, Office of Basic Energy Sciences, through the Integrated Diffraction Analysis Technologies (IDAT) program, supported by DOE Office of Biological and Environmental Research.

Author contributions—N. M., T. J. H., M. H., G. L. H., Z.-Q. F., and B. V. G. writing—original draft; N. M., C. D. G., T. J. H., M. H., G. L. H., Z.-Q. F., and B. V. G. validation; N. M., C. D. G., T. J. H., M. H., G. L. H., Z.-Q. F., and B. V. G. methodology; N. M., C. D. G., T. J. H., M. H., G. L. H., Z.-Q. F., and B. V. G. investigation; N. M., C. D. G., T. J. H., M. H., G. L. H., Z.-Q. F., and B. V. G. conceptualization; B. V. G. writing—review and editing; B. V. G. visualization; B. V. G. supervision; B. V. G. project administration; B. V. G. funding acquisition.

Funding and additional information—This research was funded by grant R35GM140852 from the US National Institutes of Health to B. V. G. and by a Cancer Research Award from the Terry C. Johnson Center of Kansas State University to C. D. G. Additional support for the SIBYLS beamline comes from the National Institute of Health project ALS-ENABLE (P30GM124169) and a High-End Instrumentation Grant (S10OD018483). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Conflict of interest—The authors declare that they have no conflicts of interest with the contents of this article.

Abbreviations—The abbreviations used are: BMP, 1,11-bismaleimido-triethyleneglycol; CG, cathepsin-G; EAP, extracellular adherence protein domain; MSA₂PV-AMC, methoxysuccinyl-Ala-Ala-Pro-Val-7-amino-4-methylcoumarin; NE, neutrophil elastase; NETs, neutrophil extracellular traps; NSP, neutrophil serine protease; SA₂PF-pNA, succinyl-Ala-Ala-Pro-Phe-p-nitro-analide; SAXS, small angle X-ray scattering; SEC-SAXS, size-exclusion chromatography in-line with small angle X-ray scattering.

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